Expo and Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach the final defence through evidence, not polish.

Keep final inputs reserved until candidates are fixed, collect actual traces and assess individual reasoning separately from group results.

Keep one habit: Look for faithful execution, honest comparison, human control and justified limits.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Evaluate a saved candidate and a manual **baseline** fairly

This final week evaluates the saved paper candidate, not presentation polish or an imagined trained model. Fix instructions before revealing the new catalogue and requests.

Keep the key below teacher-only until **observation**s are recorded. If a pupil has already seen these cases, use a fresh teacher-checked set and record that substitution; do not call familiar cases unseen.

By the end, I can defend a project using measured results, limits and an honest **baseline** comparison.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

- Saved candidate, manual **baseline**, development folder and blank record sheets with the fields listed in W2. Provide the catalogue and requests without the answer key after candidates are fixed.

- Reserved catalogue B version 4 keeps B1 to B4 order and names, with times 18, 14, 11, 7 minutes and quiet labels Yes, No, Yes, No. All materials are assumed available. Same **input** and matching rules as before.

- Reserved requests and key: F1 7 No → B4; F2 11 Yes → B3; F3 14 No → B2; F4 18 Yes → B1; F5 6 No → no match; F6 7 Yes → no match; F7 0 No → no match; F8 30 Yes → B1; F9 31 No → clarify; F10 blank minutes Yes → clarify.

  For F1–F4, earlier rows fail time; the selected row fits with equality and satisfies quiet. F5 and F7 fail all times. F6 fits only nonquiet B4, which quiet excludes. F8 accepts first quiet B1.

  F9 and F10 fail validation before search.

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

Look for: Repair a located defect without changing the agreed requirements

Reference example, not the only acceptable wording: Expected B1 with equality and quiet satisfied; S0 incorrectly returns no match because its time condition is strictly less than. S1 uses no more than, including equality.

Preserve validation, quiet, first-match order, advancement, stopping and reset. Save S0, its failing trace and exact request/catalogue labels beside the new S1 change record and rerun.

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Look for: Plan routine cases with expected results derived from the requirements

Reference example, not the only acceptable wording: R1 B1 because time fits and it is quiet; early match. R2 B2 because B1 is too long, B2 fits and quiet is unrestricted; later match.

R3 no match because every time exceeds 6; exhaustion. R4 no match because B2 fails quiet while other rows exceed 9; both conditions. Record P1 version 1, B version 1 and the manual **baseline**.

A full pass needs the required **output** and a correct explanation.

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Turn the project outcome into precise **input** and **output** requirements

Reference example, not the only acceptable wording: Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first compatible activity in B1 to B4 order.

No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

No. It informed the change and is now development **evidence** for the revision. Keep the old candidate and its final result; a fresh evaluation of the revision needs new reserved cases.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Improve One Cause

Test Normal Cases

Choose Inputs and Outputs

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

Concept to explain

Understanding is shown by explanation, **transfer**, testing, and limits.

Term	Meaning
demonstration	live showing of system behavior
defense	evidence-based explanation under questions
transfer	applying understanding to a fresh case
limitation	condition where performance or use is restricted

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked **demonstration** 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach final evaluation as an honest, fixed-candidate comparison

> Keep the eight-attempt illustration, ten-attempt exhibition example and actual final records distinct.

Save the candidate and apply the unchanged contract

This week you will evaluate your activity adviser and explain what its **evidence** supports. A project with an honestly reported failure can show strong understanding.

Save your candidate instructions, version label, **input** rules, development records and known problems before your teacher reveals the final requests and catalogue. Keep the instructions unchanged during this evaluation.

The agreed rule remains: validate whole available minutes from 0 to 30 and Quiet required Yes or No. No allows quiet or nonquiet activities.

Start at the first row and suggest the first activity whose time is no more than available time AND whose quiet label satisfies the request. Explain the choice. If all rows fail, report no match.

Invalid or missing **input** needs clarification before searching. Clear the previous **output** for each run.

Your teacher will supply a fresh, labelled catalogue in the same format. Its values are test **input**s, not permission to rewrite your instructions. All materials are assumed available.

The manual **baseline** checks every row against the same requirements, then selects the first fit. Both methods advise only; neither books, publishes or chooses on a person's behalf.

Record both methods before revealing the key

Use the same final requests, catalogue, requirements and full-pass rule for both methods. Record the candidate **output** and reason before looking at the expected answer.

Record the **baseline** separately; do not copy the candidate result into it. Your teacher holds the answer key.

A full pass needs both the required **output** and a valid reason, including why earlier rows fail when a later row is selected.

For each request record its ID, **input**s, catalogue label, candidate label, candidate **output** and reason, **baseline output** and reason, and both results after checking. Keep a trace for any failure. Mark unrun cases explicitly.

Do not silently correct an **observation**. If a person misreads an instruction, record that execution error separately from a defect in the written instruction.

Model the separate eight-attempt illustration

Here is a fictional example from a different project, not your class results. A candidate has 6 full passes from 8 attempted cases; the **baseline** has 8 from the same 8. Two more planned cases were not run.

The candidate passed 75 percent of attempted cases; the **baseline** passed 100 percent. Neither has **evidence** for the two unrun cases. The candidate did not outperform the **baseline** on this measure.

Evidence	Supported statement	Claim to avoid
6 of 8 full passes	Six attempted cases met the full rule	It works for every request
Both methods pass the same cases	They tie on this measured result	Automation must be better
A time was not measured	Speed is unknown	The candidate is faster

Evidence	Supported statement	Claim to avoid
Only fictional catalogue data	Results concern these test conditions	People will enjoy every suggestion

Explain each failure and its consequence instead of hiding it inside a total. Even perfect results on a small reserved set do not establish reliability in every situation.

Our paper adviser applies human-**written rule**s; it has not learned a model from **training** examples. It checks compatibility, not enjoyment, popularity or real material availability.

> **In the fictional example, state the candidate and baseline results, explain how the two unrun cases should be reported, and give one useful next action. Why is "the adviser is ready for every classroom" unsupported?**

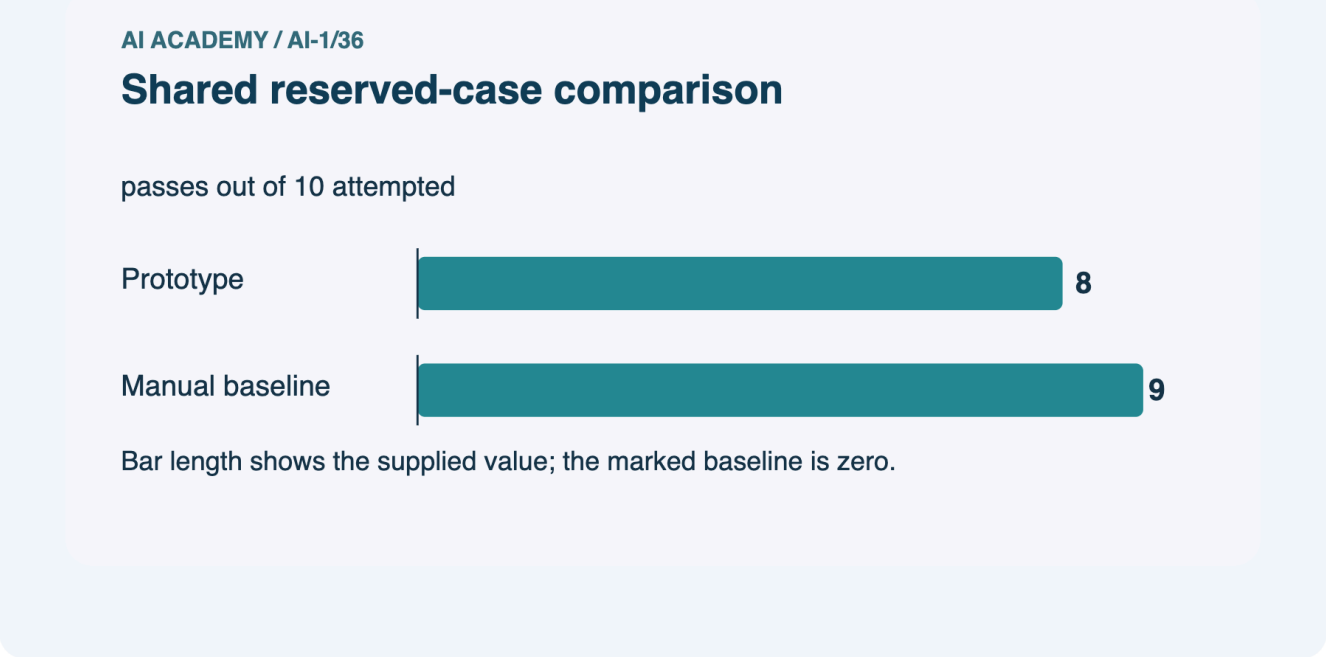
Introduce the separate exhibition-board record

A fictional paper sorter team prepares an exhibition **evidence** board.

Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

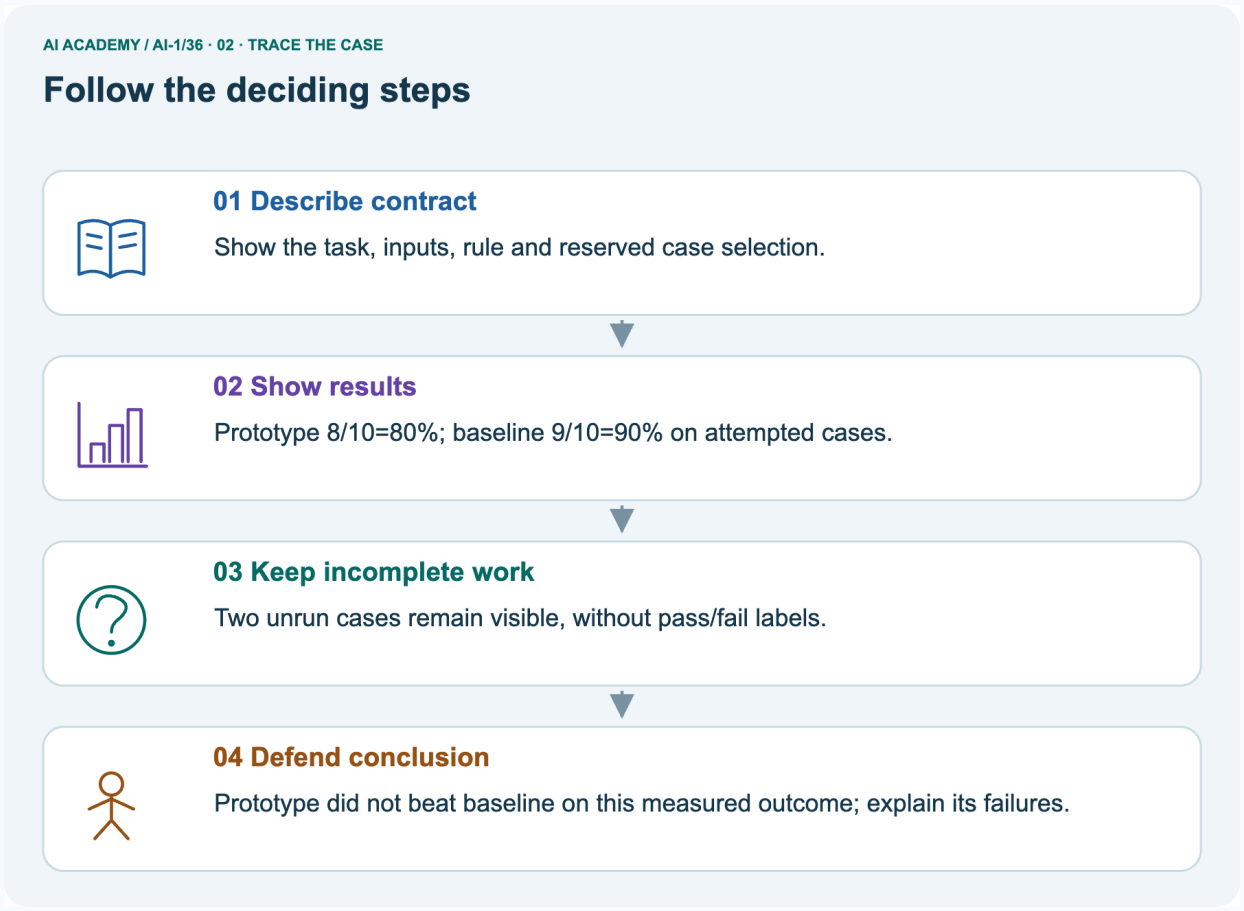
Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

Inspect the shared-case count picture



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the ten-attempt comparison and its limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Describe contract	Show the task, inputs, rule and reserved case selection.
2. Show results	Prototype 8/10=80%; baseline 9/10=90% on attempted cases.
3. Keep incomplete work	Two unrun cases remain visible, without pass/fail labels.
4. Defend conclusion	Prototype did not beat baseline on this measured outcome; explain its failures.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Keep learning-impact claims bounded

This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Distinguish planned completion from attempted-case correctness

Change: Someone adds the two unrun cases to the denominator as failures.

Prediction: That reports a different measure, 8/12 completed successes out of planned cases.

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Use the picture to request actual supporting evidence

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT fresh item and documented system

2. Process — the deciding step

PROCESS execute and explain each stage under questions

3. Output — the result

OUTPUT result, **evidence**, **limitation**, and responsible next action

CORE IDEA Understanding is shown by explanation, **transfer**, testing, and limits.

UNDER THE HOOD A **defense** evaluates **transfer**able understanding, not presentation polish.
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Strong **evidence** includes system architecture, version history, planned tests, failures, controlled improvement, fairness/privacy boundaries, and independent response to a novel case.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A team presents the Classroom Sorter, traces a new item live, shows test **evidence**, explains one failure, and states when a human must decide.

STEP 1 — OBSERVE fresh item and documented system

STEP 2 — TRACE execute and explain each stage under questions

STEP 3 — CONCLUDE result, **evidence**, **limitation**, and responsible next action

MODEL REASONING A complete **defense** traces **input**, **process**, and **output**; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

DO NOT OVERCLAIM Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

What new **evidence** could change this answer?

Concrete teaching case

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Exact explanation to model

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

Explain the added worked example

Key terms

Term	Meaning
Final evaluation	Testing a saved candidate under stated conditions after development

Term	Meaning
Baseline	A comparison method tested under the same conditions
Evidence	Recorded observations that support or challenge a claim
Limitation	A boundary on what a result or system can establish
Defence	An explanation of a decision using evidence and reasoning

Model the reasoning aloud

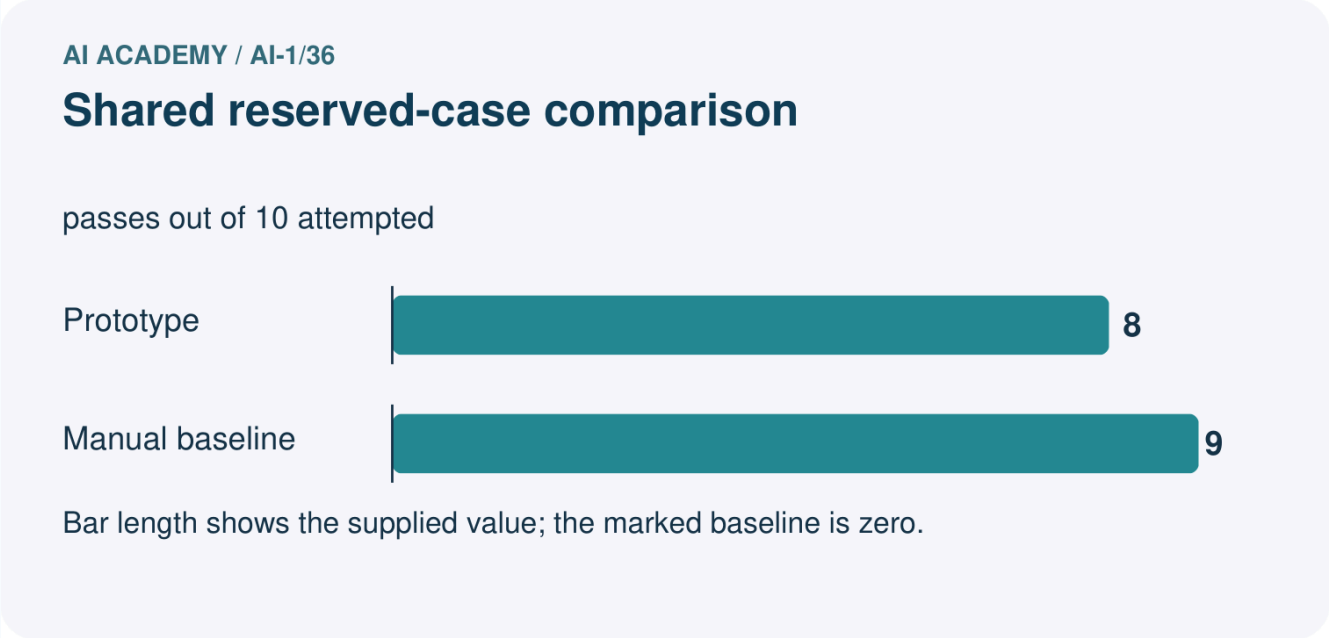
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

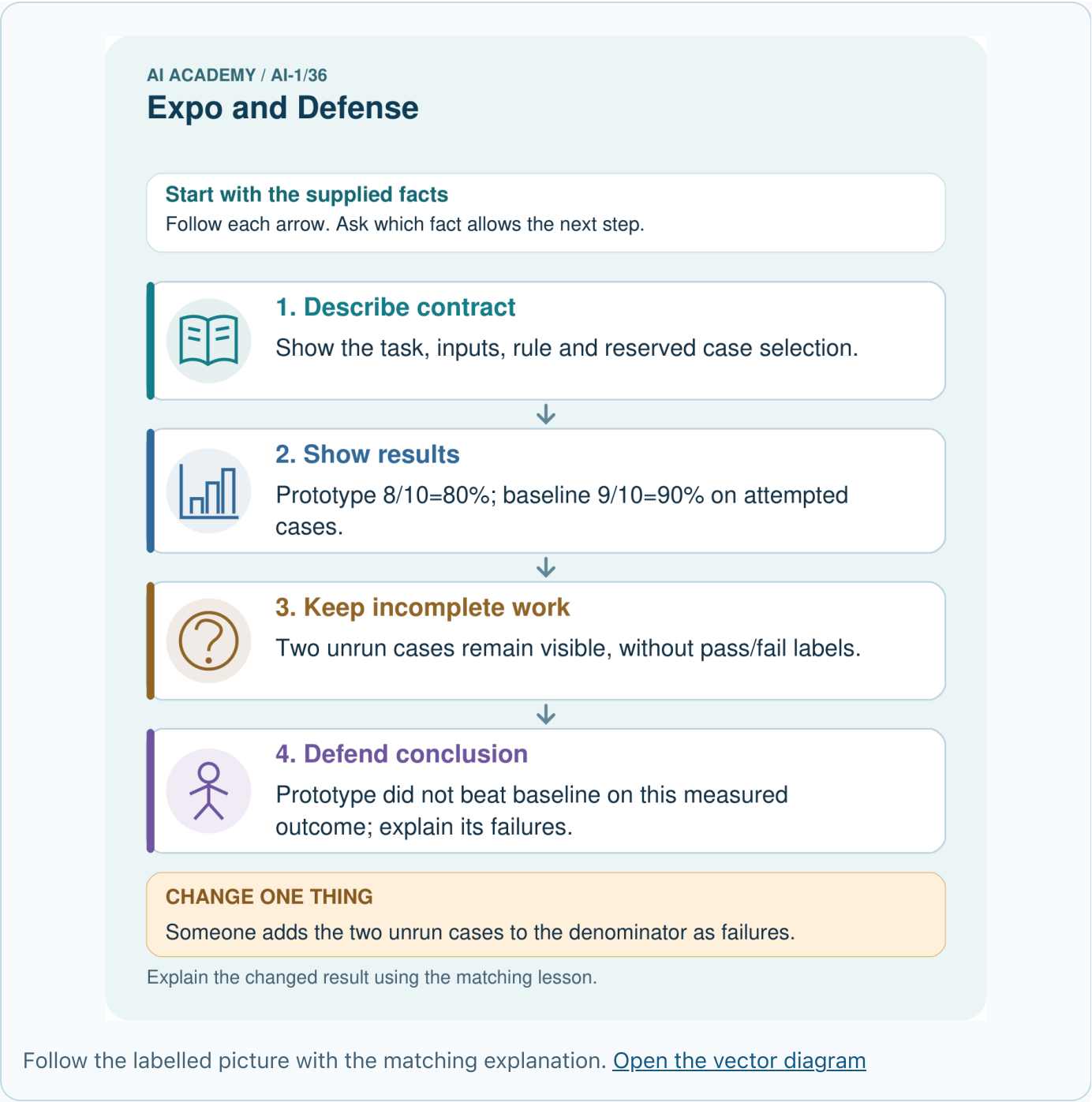
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Expo and Defense

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Use a separate sheet for the teacher's reserved cases. For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking.

Preserve failure traces. Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

Expected reasoning and fading help

Teacher check: Use the reserved key in Preparation only after **observation**s are saved. Full pass requires **output** and reason; no-match explanations must rule out all rows. Clarification must precede search.

Report actual candidate and **baseline** counts over the same attempted cases, plus unrun cases. Never substitute the key for an **observation** or assume a **baseline** run was correct.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Evaluating and explaining your project. Proposed explanation: "A polished presentation proves the system works". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the saved instruction, trace and actual result behind the claim.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Write one defensible success statement and one claim the **evidence** cannot support.

Defensible: 8 of 10 attempted reserved cases passed. Unsupported: the prototype is faster or more accurate than the **baseline**.

Changed-case reasoning: That reports a different measure, 8/12 completed successes out of planned cases. It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: The expo poster says always correct after six classroom passes.

Write the question to explain: Write a defensible claim.

Reveal after **prediction**s: Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

Use the earlier changed question as optional later practice. Reserve Judge a different project independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Fictional candidate 6/8 and **baseline** 8/8 attempted, with two unrun planned cases each; candidate 75 percent and **baseline** 100 percent on that set. Investigate retained failure traces before a versioned repair.

Neither has **evidence** for all classrooms. Exit answers must use the pupil's real record, limit the conclusion, name a next step and distinguish explicit rules from fitted models. Passing tests grants no booking authority.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Check the prerequisite and W1; collect or photograph each fixed instruction set before revealing any final inputs.
7 to 12 minutes	Reveal B version 4 and F1–F10 requests without the key. Explain W2 record fields and the unchanged full-pass criterion.
12 to 37 minutes	Run all ten cases with candidate and baseline. Partners swap operating and recording; preserve each output before comparison with the key. Record assistance and unrun cases.

Stage	Teacher action
37 to 45 minutes	Reveal the key, check reasons and summarise actual results in W2. Retain failure traces; do not edit the evaluated instructions.
Session 2 start to 8 minutes	Discuss the fictional 6-of-8 example and collect the guided explanation. Keep hypothetical results separate from class records.
8 to 22 minutes	Prepare W3 and run paired two-minute presentations with questions and role swaps. Sample individual traces; offer accessible spoken or written responses.
22 to 37 minutes	Collect W4 independently before hints. Assess evidence claims and the meaning of a rerun after revision.
37 to 45 minutes	Complete W5 and the exit. Archive the candidate, final results and next action; note support needs for Grade 7.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the final activity evaluation, use the reserved catalogue, requests and key in the teacher's preparation only after candidates are fixed. Keep the key private until actual candidate and baseline observations are saved. Do not fill observations from expected answers or assume the baseline is error-free. Full pass needs output plus supported reason, with earlier rows ruled out for a later fit. Retain unrun cases, assistance, execution errors, instruction defects

and original results separately. Presentation should use actual evidence and allow individual oral or written explanations. If learners have already seen the reserved set, use a new teacher-checked set and record the substitution. Reserve Judge a different project independently for the individual assessment.

Run and record

1. Complete the task: Demo the system, solve a fresh case, and answer why it may fail.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one **limitation**, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A polished presentation proves the system works: Ask for the saved instruction, trace and actual result behind the claim.

Unrun cases count as passes: Report attempted denominators and unrun cases separately.

The **baseline** must lose: A tie or a better **baseline** can justify retaining the simpler method.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Judge a different project independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR A successful **demonstration** proves universal reliability.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A repair erases the final failure: Preserve the evaluated version and results; revealed cases become development checks for the revision.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

Report candidate 4 of 6 and **baseline** 5 of 6 full passes, with one unrun case for each. The **baseline** performed better on this measured set; neither result establishes reliability everywhere. Speed was not measured.

A rerun after the failures informed a repair is development **evidence** for the revised version, not an unseen final test. Preserve the old results and label both versions.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Record the fixed candidate

Before final cases are revealed, record your version label, where its complete instructions are saved, the agreed **input** and matching rules, the manual **baseline** and any known **limitation**.

Confirm that the instructions stay unchanged while the assigned catalogue is used as **input**.

Require an identifiable saved version, complete instructions, valid whole minutes 0 to 30, Yes or No quiet **input**, inclusive time AND quiet compatibility, first-fit order, reset, stopping and clarification.

The **baseline** checks all rows and selects the first fit under the same contract. Accept honestly stated known failures; do not award a stronger result for hiding them.

W2 Keep the final evaluation record

Use a separate sheet for the teacher's reserved cases. For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking. Preserve failure traces.

Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

Use the reserved key in Preparation only after **observation**s are saved. Full pass requires **output** and reason; no-match explanations must rule out all rows. Clarification must precede search.

Report actual candidate and **baseline** counts over the same attempted cases, plus unrun cases. Never substitute the key for an **observation** or assume a **baseline** run was correct.

W3 Prepare your evidence defence

Write a two-minute explanation of the problem, one trace, your actual comparison, a **limitation** and a next action. Explain why this is a written-rule design rather than a model trained on examples.

State which personal information is unnecessary and who decides whether to act.

Look for a saved trace connected to the actual instruction, accurate counts, a specific failure or **limitation** and a justified next step. The project applies explicit human-**written rule**s, with no fitted model.

Only minutes and quiet preference are needed; names and reasons for quiet are unnecessary. The person may decline and the teacher checks real conditions; the adviser does not book or publish.

W5 Write your final conclusion

Use your own records to complete: Under these conditions, our saved version showed… The **baseline** showed… We still do not know… Our next action is… Include one privacy or human-control choice.

If a case failed, explain its consequence without deleting the result.

Accept different conclusions grounded in actual **observation**s, including a decision to retain the **baseline**. Require stated conditions, honest comparison, a real uncertainty and a specific action.

A quiet requirement does not require a diagnosis; an **output** does not authorise action. A failure remains part of the evaluated version’s record even after a repair.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

Additional transfer practice question

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

Reasoned practice answer

Say it succeeded on these four named cards, with the observed count reported. Four cases do not prove it never fails. Next test unfamiliar, blurred or dim examples and explain human review.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A successful **demonstration** proves universal reliability.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Overall is 12/16=75%; routine is about 83.3%; mixed is 50%. Show the actual failed trace and investigate mixed-material handling or review rules. 'Works reliably on every material' overstates these small specified tests.

Advancement requires independent explanation, not presentation polish alone.

Explanations for the additional workbook examples

W1 Record the fixed candidate

W2 Keep the final evaluation record

W3 Prepare your evidence defence

W4 Judge a different project independently

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

Report candidate 4 of 6 and **baseline** 5 of 6 full passes, with one unrun case for each. The **baseline** performed better on this measured set; neither result establishes reliability everywhere. Speed was not measured.

A rerun after the failures informed a repair is development **evidence** for the revised version, not an unseen final test. Preserve the old results and label both versions.

W5 Write your final conclusion

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

Keep the decision, **input**, result, explanation and remaining **limitation** in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Provide a blank record grid, read instructions aloud and allow oral **defence**, while recording assistance. Keep independent W4 responses before feedback.

Allow additional time for unrun cases with the same fixed version and record when they ran; do not manufacture completion.

Extension: propose an additional test for a stated **limitation** and explain why it would help without claiming proof of every situation.

Assess four strands separately: faithful execution with traces; honest comparison including failures; independent explanation of rule-based **mechanism** and **human control**; justified limits and next action.

Mark each secure, developing or not yet **evidence**d, with one cited **observation**. Do not infer individual mastery from group totals. W4 requires 4/6 versus 5/6, one unrun, unknown speed and a development rerun.

After support, retry with candidate 3/5 and **baseline** 3/5, two unrun and no timing: a tie on observed passes, no universal or speed claim. Require fresh reasoning.

Plan the delayed check

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Review the full project **evidence** and choose concepts needing another independent attempt before the next level. One expo is not proof of mastery of all AI.

Grade 6 Foundations is complete. Grade 7 AI 2 begins programming and data. Carry forward versioned instructions, validation, tests, privacy and **evidence**-based conclusions; use the recorded assessment strands to plan support.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Evaluate a saved candidate and a manual **baseline** fairly

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Use a separate sheet for the teacher's reserved cases.

For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking. Preserve failure traces.

Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

Reasoned practice answer: Use the reserved key in Preparation only after **observation**s are saved. Full pass requires **output** and reason; no-match explanations must rule out all rows. Clarification must precede search.

Report actual candidate and **baseline** counts over the same attempted cases, plus unrun cases. Never substitute the key for an **observation** or assume a **baseline** run was correct.

Check every learner before discussion: Use the worked case for Evaluating and explaining your project. Proposed explanation: "A polished presentation proves the system works". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the saved instruction, trace and actual result behind the claim.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Someone adds the two unrun cases to the denominator as failures. Require a **prediction** and a reason before comparing. Expected result: That reports a different measure, 8/12 completed successes out of planned cases.

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ Reference: feedback and records

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

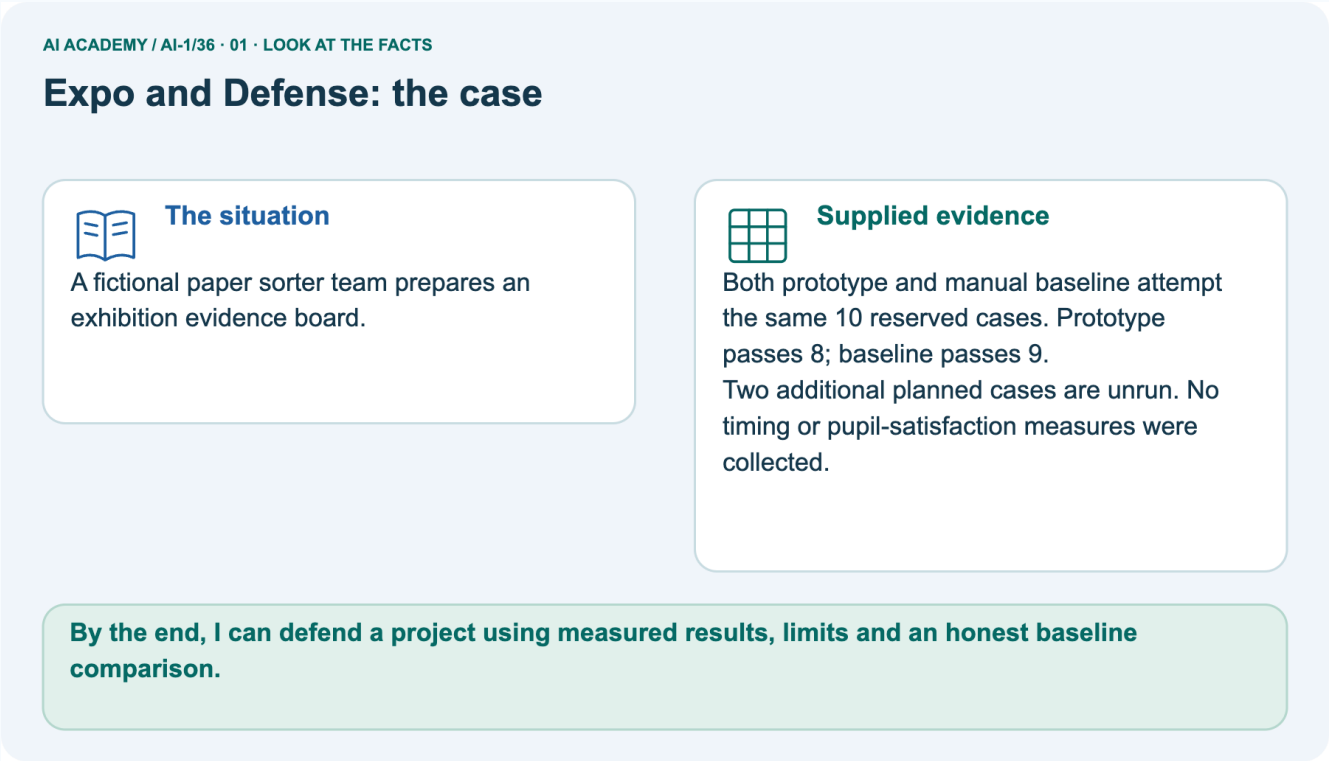
Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

▼ Reference: teach with the six pictures

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Expo and Defense: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

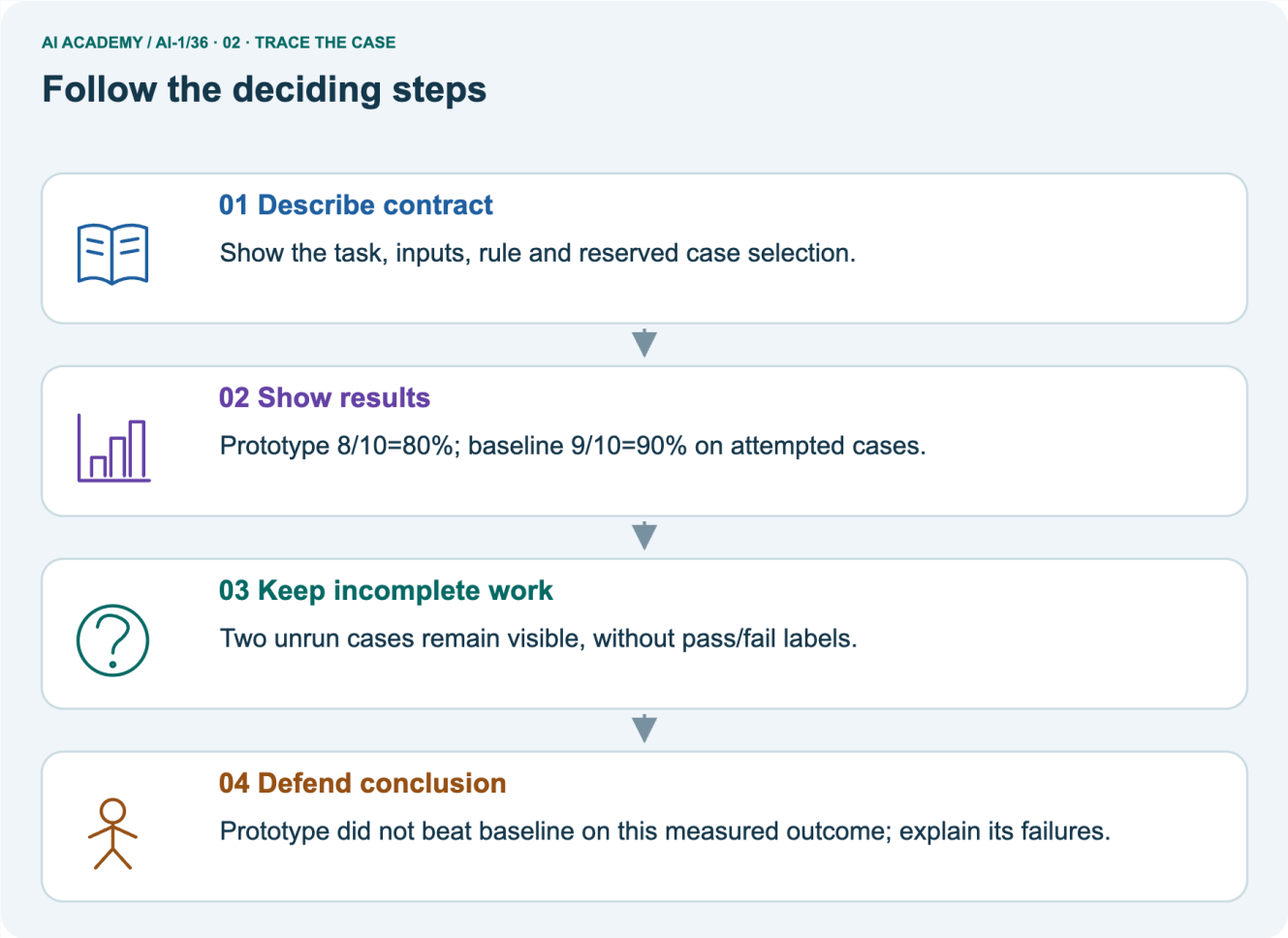
A fictional paper sorter team prepares an exhibition **evidence** board. Supplied facts: Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



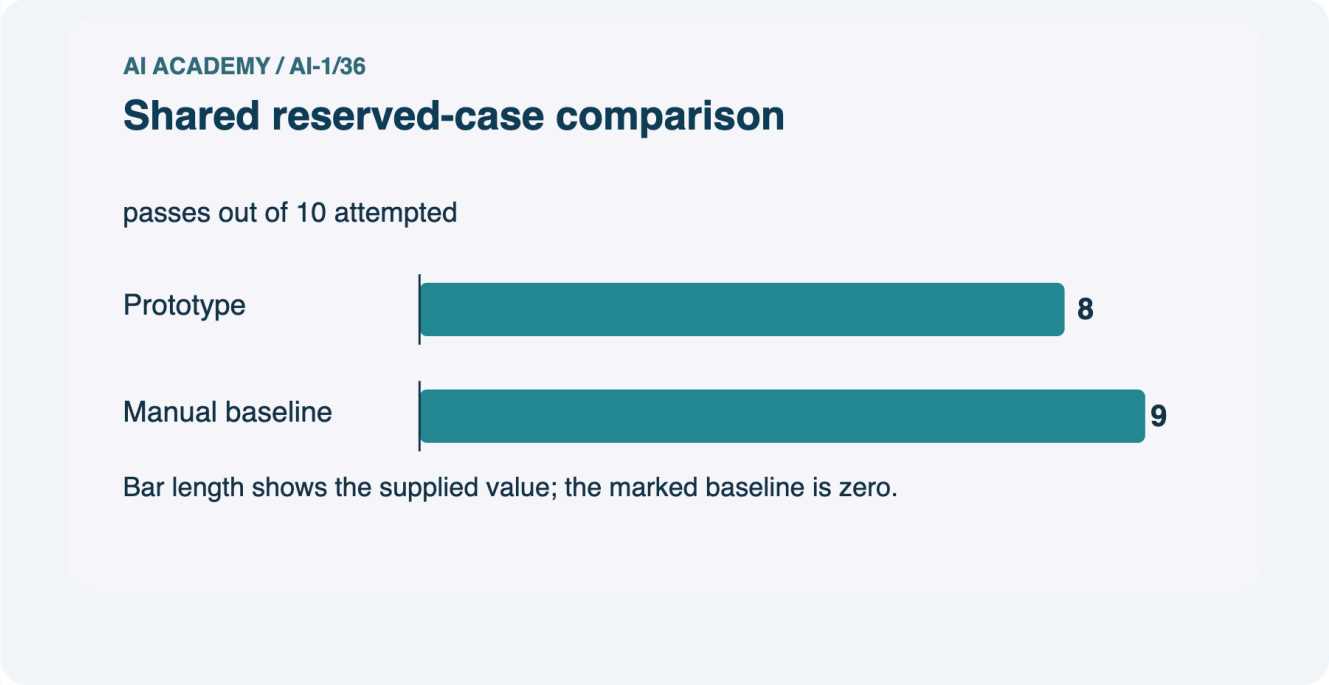
Follow the labelled picture with the matching explanation. Open the vector diagram

Describe contract: Show the task, **input**s, rule and reserved case selection. → Show results: Prototype 8/10=**80%**; **baseline** 9/10=90% on attempted cases. → Keep incomplete work: Two unrun cases remain visible, without pass/fail labels. → Defend conclusion: Prototype did not beat **baseline** on this measured outcome; explain its failures.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Shared reserved-case comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

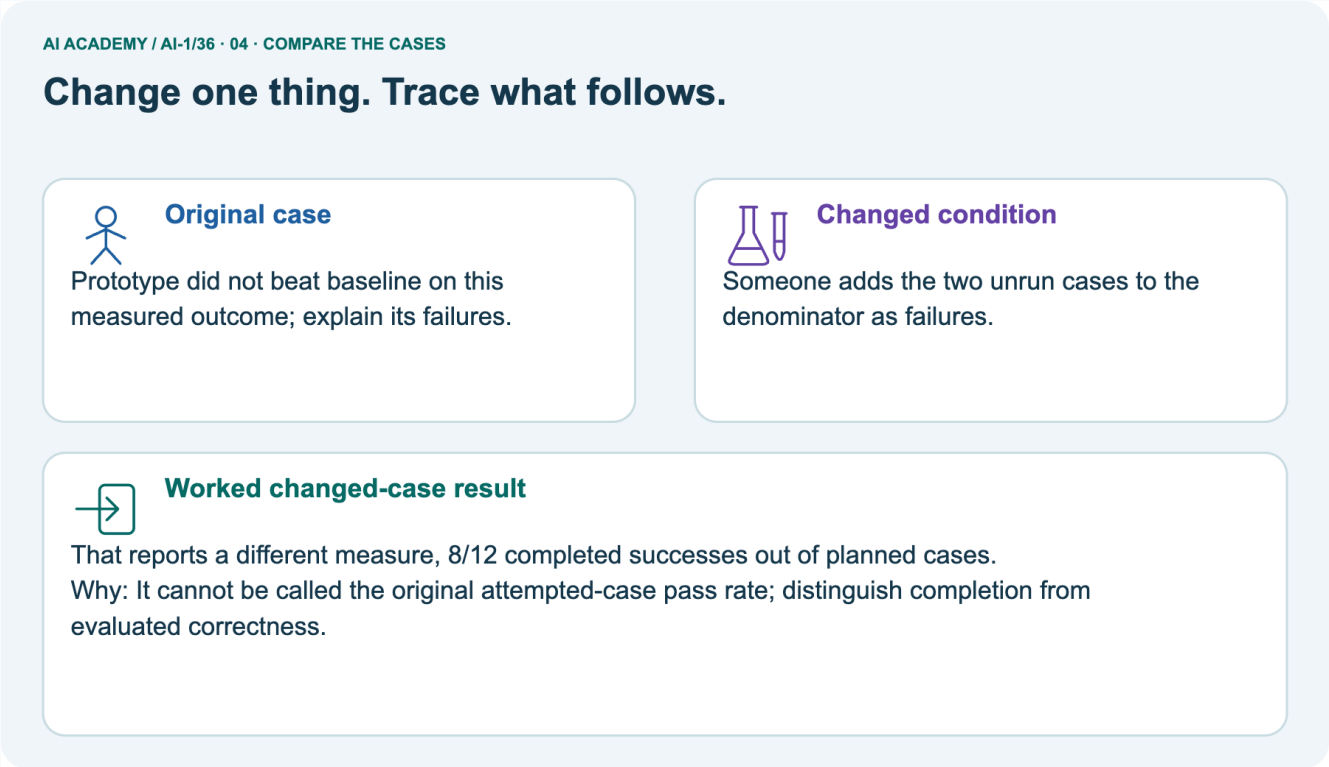
Shared reserved-case comparison. Units: passes out of 10 attempted. Prototype: 8. Manual **baseline**: 9.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Prototype did not beat **baseline** on this measured outcome; explain its failures. Changed condition: Someone adds the two unrun cases to the denominator as failures.

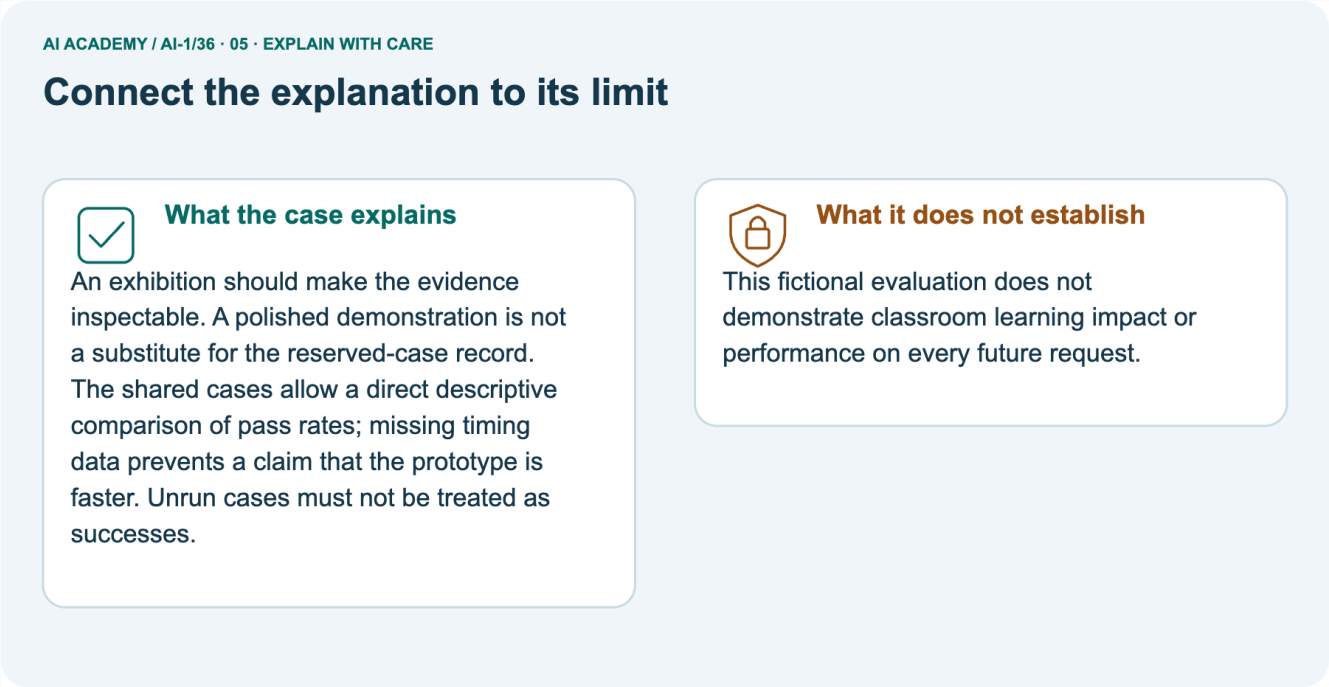
Expected result: That reports a different measure, 8/12 completed successes out of planned cases. Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

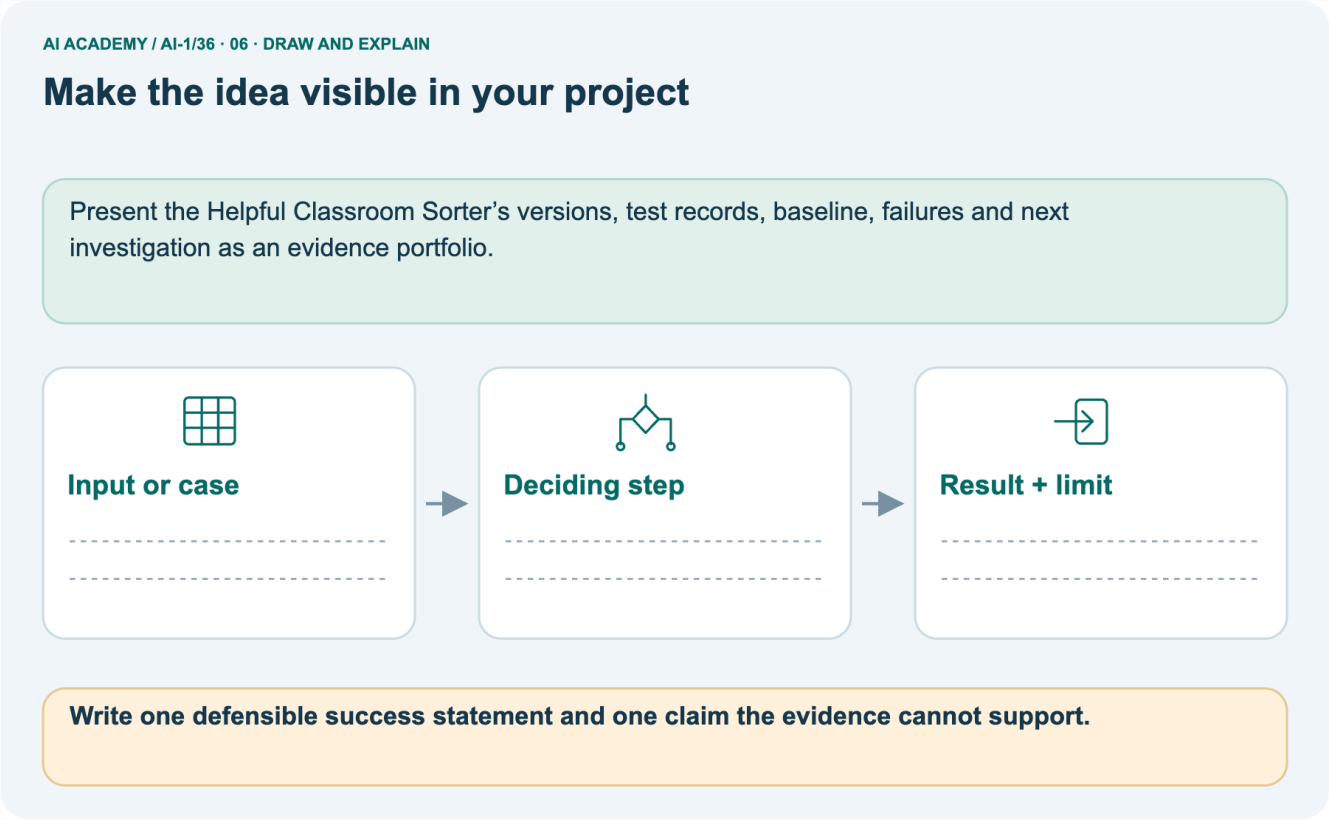
Limit: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio. Draw your **input**, deciding step, result and limit.

Supported question: Write one defensible success statement and one claim the **evidence** cannot support.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Defensible: 8 of 10 attempted reserved cases passed. Unsupported: the prototype is faster or more accurate than the **baseline**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Expo and Defense: the case

02. Follow the deciding steps

03. Shared reserved-case comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.